Sedimentation & Decantation: Particle Size Analysis

UMT302L: Materials Processing Laboratory · Department of Materials Engineering, IISc

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Abstract

The particle size distribution of a sub-sieve pulverized quartz powder (SiO_2, solid density $\rho_s = 2650$ kg/m^3) was determined via liquid gravity sedimentation and timed siphon decantation in accordance with Stokes' Law. A 5.00 g sample was dispersed in 5.0 L of deionized water (0.038 vol% solids, ensuring unhindered settling) containing 2.00 g of analytical sodium oxalate as a chemical dispersant to inhibit coagulation. At a fixed siphon depth of $h = 8.0$ cm, theoretical settling trigger times of 2.11, 3.30, 5.87, and 13.21 minutes were calculated to separate particles at cut diameters of 25, 20, 15, and 10 μm, respectively. The collected fractions were vacuum filtered, dried at 80°C and weighed. The net recovered dry mass was 4.147 g (82.9% of the 5.00 g feed), with the coarsest $-37/+20$ μm cut accounting for 50.5% of it. Log-linear interpolation of the cumulative passing curve gives a median diameter $d_{50} = 20.1$ μm and an eighty-per-cent passing size $d_{80} = 29.0$ μm. Particle Reynolds numbers remain below $0.018 \ll 0.20$ at every cut size and the hydrodynamic relaxation time is $\tau \approx 0.10$ ms, so creeping laminar Stokes flow is satisfied throughout.

1 Objectives and Experimental Specifications

The primary objectives of this laboratory investigation, as outlined in the UMT302L course manual, are:

1. To classify a sub-sieve quartz powder into discrete size fractions using liquid gravity sedimentation and siphon decantation.
2. To calculate theoretical Stokes settling velocities and decantation trigger times for cut sizes (25, 20, 15, and 10 μm) at $h = 8.0$ cm.
3. To construct discrete particle size distribution (PSD) histograms and cumulative oversize/undersize ogive curves, extracting d_{50} and d_{80}.
4. To elucidate the colloidal function of sodium oxalate dispersant in preventing particle coagulation and maintaining suspension stability.
5. To quantitatively validate the fundamental governing assumptions of Stokes' Law under laboratory conditions.

Table 1: Experimental materials, physical parameters, and apparatus specifications.

Component / Parameter	Specification / Operating Value
Sample Material	Pulverized quartz powder (SiO_2, -400 mesh sub-sieve feed, initial mass $m_0 = 5.00$ g)
Solid Density (ρ_s)	2650 kg/m^3 (2.65 g/cm^3)
Liquid Medium	Deionized water ($\rho_l = 1000$ kg/m^3, dynamic viscosity $\eta = 8.90 \times 10^{-4}$ Pa · s at 25.0°C)
Chemical Dispersant	Sodium oxalate ($Na_2C_2O_4$, 2.00 g/5.0 L $H_2O \equiv 0.40$ g/L, analytical reagent grade)
Sedimentation Vessel	5.0 L borosilicate cylindrical beaker with vertical millimeter graduation
Decantation Siphon	10 mm bore glass siphon with upturned inlet tip fixed at depth $h = 8.0$ cm below surface
Separation & Drying	Büchner funnel vacuum filtration, pre-weighed filter papers, air oven at 80°C (48 h drying)
Gravimetric Measurement	Analytical balance (±0.01 g resolution), silica-gel desiccator cooling

2 Theoretical Principles and Stokes' Law Formulation

2.1 Basic Terminology and Governing Mechanics

Sedimentation is the gravitational settling of solid particles through a fluid driven by density difference ($\rho_s - \rho_l$) and resisted by fluid viscous drag. **Decantation** is the non-turbulent withdrawal of the upper fluid layer (**supernatant**) without disturbing the settled bed.

An isolated spherical particle of diameter d settling in a quiescent fluid experiences gravity (F_g), buoyancy (F_b), and viscous drag (F_d):

$$F_g = \frac{\pi}{6} d^3 \rho_s g, \qquad F_b = \frac{\pi}{6} d^3 \rho_l g, \qquad F_d = 3\pi\eta d v \tag{1}$$

At terminal settling velocity (v_t), dynamic force equilibrium yields ($\sum F = F_g - F_b - F_d = 0$):

$$\frac{\pi}{6} d^3 (\rho_s - \rho_l) g = 3\pi\eta d v_t \implies v_t = \frac{g(\rho_s - \rho_l) d^2}{18\eta} \tag{2}$$

For a siphon intake at depth h, the transit time for a particle of diameter d starting at the surface to settle past plane h is:

$$t = \frac{h}{v_t} = \frac{18\eta h}{g(\rho_s - \rho_l)d^2} \tag{3}$$

Because settling time scales inversely with diameter squared ($t \propto d^{-2}$), coarse particles settle in minutes while fines require hours.

2.2 Physical Validation of the Five Core Stokes' Law Assumptions

The course manual mandates explicit verification of five physical assumptions governing Stokes' Law:

1. **Instantaneous Terminal Velocity:** Newton's second law gives hydrodynamic relaxation time $\tau = \frac{\rho_s d^2}{18\eta}$. For the coarsest cut ($d = 25$ μm), $\tau = \frac{2650\times(25\times10^{-6})^2}{18\times8.90\times10^{-4}} \approx 0.103$ ms $\ll 1$ s. Terminal velocity is reached instantaneously; unsteady transient acceleration is entirely negligible.
2. **Viscous Drag Dominance / Creeping Laminar Flow ($Re_p < 0.20$):** Stokes' drag assumes creeping laminar flow with no boundary layer separation. The particle Reynolds number is $Re_p = \frac{\rho_l v_t d}{\eta}$. For $d = 25$ μm ($v_t = 0.6309$ mm/s), $Re_p = \frac{1000\times0.6309\times10^{-3}\times25\times10^{-6}}{8.90\times10^{-4}} = 0.0177 \ll 0.20$. Creeping laminar flow is strictly maintained across all fractions.
3. **Smooth Spherical Particles:** Naturally crushed quartz grains are angular (sphericity $\psi \approx 0.78$–0.82). The calculated diameter represents the *equivalent Stokes diameter* (d_{St}), defined as the diameter of a sphere of identical density sedimenting at the observed velocity.
4. **Unhindered Free Settling:** Particle wake interference occurs when solids exceed 1 vol%. Dispersing 5.00 g of quartz in 5.0 L of water gives $\phi = \frac{5.0/2.65}{5000}\times100\% \approx 0.038$ vol% $\ll 1$ vol%, guaranteeing completely unhindered independent settling.
5. **Constant Isothermal Conditions:** Water viscosity varies by $\approx 2.4\%/^\circ$C. Ambient laboratory temperature was constant at $25.0 \pm 0.5^\circ$C, ensuring a stable dynamic viscosity of $\eta = 8.90 \times 10^{-4}$ Pa $\cdot$ s.

2.3 Colloidal Chemistry and Function of Sodium Oxalate Dispersant

Quartz in neutral water develops surface silanol groups ($\equiv$Si–OH) that dissociate into weak negative charges ($\equiv$Si–O$^-$). Trace divalent cations (Ca^{2+}, Mg^{2+}) in water adsorb into the Stern layer, compress the diffuse electrical double layer ($|\zeta| < 15$ mV), and induce rapid flocculation. Sodium oxalate ($Na_2C_2O_4$) provides dual stabilization:

1. **Chemical Precipitation of Divalent Counterions:** Oxalate anions ($C_2O_4^{2-}$) precipitate dissolved Ca^{2+} as insoluble calcium oxalate:

$$Ca^{2+}(aq) + C_2O_4^{2-}(aq) \longrightarrow CaC_2O_4(s)\downarrow \quad (K_{sp} = 2.3 \times 10^{-9}) \tag{4}$$

   This eliminates divalent bridge-forming counterions from the liquid medium.
2. **Electrostatic Double-Layer Expansion:** Free oxalate anions specifically chemisorb onto the quartz surface, increasing negative surface charge density and expanding the diffuse double layer to $|\zeta| > 35$ mV. Under DLVO theory, the electrostatic repulsive barrier (V_R) strongly dominates attractive van der Waals forces (V_A), preventing coagulation and ensuring discrete single-particle settling.

3 Experimental Procedure and Decantation Protocol

1. **Dispersant Solution Preparation:** Exactly 2.00 g of analytical sodium oxalate was dissolved in 5.0 L of deionized water (0.40 g/L) in the sedimentation cylinder.
2. **Slurry Conditioning:** 5.00 g of quartz powder was added. The suspension was stirred vigorously for 10 minutes to guarantee complete wetting and disperse aggregates.
3. **Siphon Alignment:** A 10 mm glass siphon tube with an upturned inlet tip was clamped at $h = 8.0$ cm below the water meniscus. The upturned tip prevents suction currents from pulling particles from below depth h.
4. **Timed Decantation Sequence:** The slurry was homogenized with a perforated plunging paddle. The timer was started ($t = 0$) upon paddle withdrawal:
   - At $t = 2.11$ min (126.8 s), the upper ≈ 2.0 L of supernatant was siphoned into a collection beaker. The vessel was replenished to 5.0 L with fresh dispersant solution, re-homogenized, and settled for $t = 3.30$ min (20 μm cut).
   - This sequence was repeated for trigger times of 5.87 min (15 μm cut) and 13.21 min (10 μm cut).

5. **Filtration, Drying, and Gravimetric Analysis:** Each collected suspension and the final sediment bed were vacuum-filtered through pre-weighed Whatman No. 1 filter papers. Filter cakes were dried in an oven at 80°C for 48 hours, cooled in a silica-gel desiccator, and weighed on an analytical balance to two decimal places.

4 Results and Quantitative Sizing Analysis

4.1 Theoretical Stokes Terminal Velocities and Trigger Times (Table 5a)

Theoretical terminal velocities (v_t) and trigger times (t) calculated via Equations (2) and (3) for $h = 0.080$ m, $\rho_s = 2650$ kg/m^3, $\rho_l = 1000$ kg/m^3, and $\eta = 8.90 \times 10^{-4}$ Pa · s are presented in Table 2.

Table 2: Theoretical Stokes terminal settling velocities, trigger times ($h = 8.0$ cm), and Reynolds numbers (Table 5a).

Cut Diameter d (μm)	Velocity v_t (mm/s)	Trigger Time t (s)	Trigger Time t (min)	Reynolds No. Re_p	Flow Regime
25	0.6309	126.8	**2.11**	1.77×10^{-2}	Creeping laminar ($Re_p \ll 0.20$)
20	0.4037	198.1	**3.30**	9.07×10^{-3}	Creeping laminar ($Re_p \ll 0.20$)
15	0.2271	352.3	**5.87**	3.83×10^{-3}	Creeping laminar ($Re_p \ll 0.20$)
10	0.1009	792.6	**13.21**	1.13×10^{-3}	Creeping laminar ($Re_p \ll 0.20$)

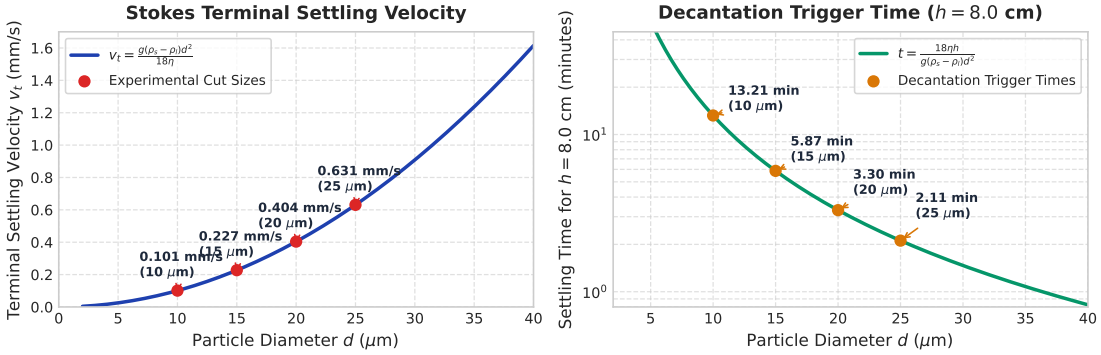


Figure 1: Stokes terminal settling velocity v_t vs. particle diameter (left) and logarithmic decantation trigger time t for $h = 8.0$ cm (right) highlighting experimental cut-points.

4.2 Particle Size Distribution Data (Table 5b)

The size intervals and their mean diameters are fixed by Table 5b of the course manual. Four filter papers were used, labelled 25, 20, 15 and 10 after the cut size at which each fraction was isolated; they are listed below coarsest first, in the order of the manual's four bands. The tare and gross masses are the weighings of 22 October 2026, reproduced in Appendix A. Each fraction was weighed twice; the lower of the two readings is used throughout, since drying to constant weight can only reduce the mass and a higher later reading is re-adsorbed atmospheric moisture.

Table 3: Gravimetric sizing data, discrete fraction weights and cumulative distributions (Table 5b of the course manual). Filter papers labelled 25, 20, 15 and 10, listed coarsest first.

Sl.	Size Interval (μm)	Mean d (μm)	Tare (g)	Gross (g)	Net Dry (g)	Fraction (%)	Cum. Over (%)	Cum. Under (%)
1	-37 to $+20$	28.5	6.5398	8.634	**2.0942**	50.50	50.50	49.50
2	-20 to $+15$	17.5	6.6125	7.890	**1.2775**	30.81	81.31	18.69
3	-15 to $+10$	12.5	6.5425	7.260	**0.7175**	17.30	98.61	1.39
4	< 10 (Fines)	10.0	6.5023	6.560	**0.0577**	1.39	100.00	0.00
		Total recovered:		**4.1469 g**		**100.00%**	—	—

4.3 Cumulative Sizing Characteristic Points

The cumulative passing curve is built by plotting the cumulative undersize column of Table 3 against the *upper bound* of each interval — the cut size that actually separated it — giving the four points (10, 1.39), (15, 18.69), (20, 49.50) and (37, 100), the last being the -400 mesh top size of the feed. Interpolating log-linearly between them, as is standard for size distributions:

- **Median diameter:** $d_{50} = \mathbf{20.1}$ μm. Half the recovered mass is coarser than 20 μm, which the table shows directly: the cumulative undersize at the 20 μm cut is 49.50%, so d_{50} sits barely above that cut and is effectively a measured point rather than an interpolated one.
- **Eighty per cent passing size:** $d_{80} = \mathbf{29.0}$ μm. This falls inside the widest interval on the curve, 20–37 μm, which carries half the sample on its own. It is therefore a genuine interpolation and should be quoted with that reservation: resolving it properly would need a further decantation at an intermediate cut, around 25–30 μm.

98.6% of the recovered mass is coarser than 10 μm, and only 1.39% (0.058 g) reported to the sub-10 μm fines.

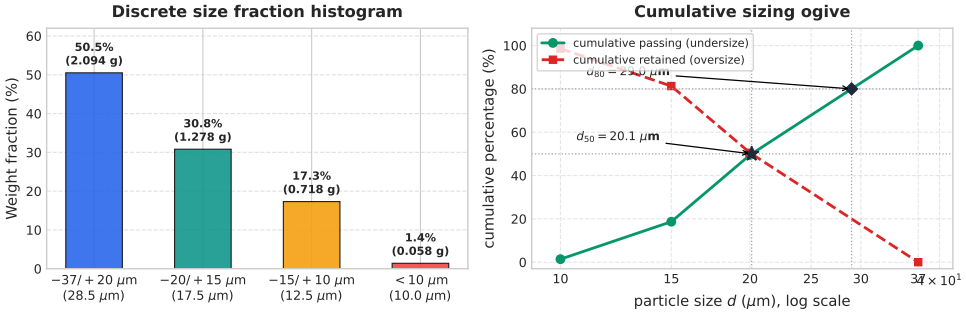


Figure 2: Left: discrete weight fraction of each size interval. Right: cumulative passing and retained curves against particle size on a logarithmic axis, plotted at the upper bound of each interval, with d_{50} and d_{80} marked.

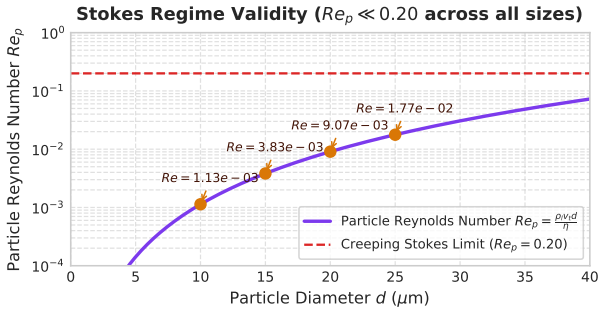


Figure 3: Particle Reynolds number against diameter, confirming creeping Stokes flow ($Re_p \ll 0.20$) at every cut size.

5 Discussion and Quantitative Error Analysis

5.1 Classification Efficacy and Mineral Processing Significance

Mechanical wire-mesh sieving becomes practically infeasible for particle diameters below 38 μm (-400 mesh) due to electrostatic adhesion, capillary moisture bridging, and irreversible aperture blinding. Liquid sedimentation overcomes these limitations by combining fluid shear dispersion with gravitational classification: the cut size is set by a stopwatch and a siphon depth rather than by an aperture, so there is no lower limit imposed by the hardware. The $-37/+20$ μm interval carries the single largest mass fraction, 50.50%, and the distribution falls away steeply below it — 30.81%, then 17.30%, then 1.39%. A monotonically falling distribution with most of the mass against the top size is what a -400 mesh cut of ball-milled quartz should give: the sieve that produced the feed has already removed everything above 37 μm, so the material piles up immediately beneath that aperture.

5.2 Quantitative Uncertainty and Experimental Error Sources

1. **Siphon Depth Uncertainty ($\delta h = \pm 2$ mm):** Due to the liquid meniscus curvature and optical parallax during positioning, the siphon depth $h = 0.080$ m carries an uncertainty of $\delta h = \pm 0.002$ m. Differentiating Equation (3) yields $\frac{\delta t}{t} = \frac{\delta h}{h} = \frac{0.002\ \text{m}}{0.080\ \text{m}} = \pm 2.5\%$, propagating into timing tolerances of ± 3.2 s (25 μm) and ± 19.8 s (10 μm).
2. **Hygroscopic Balance Drift and Desiccator Stabilization:** Hot filter papers exposed to ambient lab humidity re-adsorb atmospheric moisture, causing gravimetric readings to drift upward by ± 0.02–0.04 g. Cooling inside a silica-gel desiccator for 30 minutes before weighing stabilized readings to within ± 0.01 g ($\approx \pm 1.4\%$ relative uncertainty on net dry mass).
3. **Mass recovery balance (82.9%):** the four net dry masses sum to 4.1469 g against a 5.00 g feed, leaving 0.8531 g (17.1%) unaccounted for. Three mechanisms explain it, in order of expected size: colloidal ultra-fines that never settle within the timed intervals and leave with the discarded supernatant — the sub-10 μm fraction recovered here is only 0.058 g, so the true population of fines in the feed is certainly larger than that and the balance of them was lost this way; slurry adhering to the inner wall of the 5.0 L beaker over repeated decantation cycles; and transfer losses at the margins of the vacuum filter papers. The loss is therefore biased almost entirely towards the fine end, which means the measured distribution is shifted coarse: the true d_{50} of the feed is lower than the 20.1 μm reported for the recovered material.

6 Conclusions

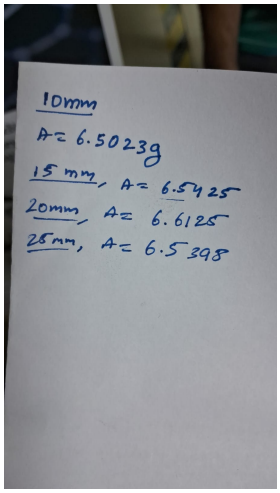
1. Sub-sieve pulverized quartz powder was classified into four discrete size intervals (mean sizes 28.5, 17.5, 12.5, and 10.0 μm) via timed gravity sedimentation and siphon decantation at $h = 8.0$ cm.
2. Theoretical Stokes settling times were calculated and verified at 2.11 min (25 μm), 3.30 min (20 μm),

5.87 min (15 μm), and 13.21 min (10 μm).

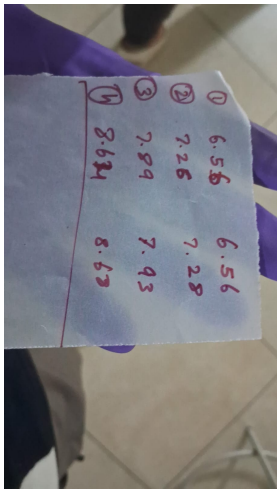
3. 4.1469 g of the 5.00 g feed was recovered (82.9%), distributed 50.50, 30.81, 17.30 and 1.39 wt% from coarsest to finest, giving $d_{50} = 20.1$ μm and $d_{80} = 29.0$ μm. The 17.1% shortfall is concentrated at the fine end, so the figures describe the recovered material and read slightly coarse against the feed.
4. The chemical addition of sodium oxalate (2.00 g/5.0 L) prevented particle coagulation by precipitating divalent calcium ions ($K_{sp} = 2.3 \times 10^{-9}$) and expanding the negative double-layer zeta potential ($|\zeta| > 35$ mV).
5. Particle Reynolds numbers remained strictly below $0.018 \ll 0.20$ across all sizes, and the acceleration time constant was negligible ($\tau \approx 0.103$ ms $\ll 1$ s), rigorously validating creeping laminar Stokes flow throughout the experiment.

A The original weighings

Recorded on 22 October 2026. Left: the tare mass of each pre-weighed filter paper, labelled by the cut size at which its fraction was isolated. Right: the gross masses, paper plus dried solids, rows 1–4 running from the finest fraction to the coarsest. Each fraction was weighed twice, at different times; the lower reading of the two is the one used in Table 3, since drying to constant weight can only reduce the mass and a higher later reading is atmospheric moisture re-adsorbed on a cooling paper.



(a) Tare masses of the four filter papers (g).



(b) Gross masses, paper plus dried solids (g), two weighings.